FP8 systolic matrix multiplier

Part II integration, verification, annotated waveforms and core synthesis comparison

Implementation. The supplied Part I FP8 multiplier, adder and round/pack RTL are reused unchanged. In the baseline 4x4 core, each integer MAC is replaced with the FP8 operations while preserving the original operand, product and accumulator registers (MAC latency parameter = 3). Multiply and add round separately.

Flags. Multiplier flags are delayed with the registered product. Each MAC accumulates delayed multiplier flags and current adder flags in a sticky register. A bitwise OR of all 16 MACs drives the new core exception_flags[4:0] output. Bit order: overflow, underflow, divide-by-zero, invalid, inexact. The existing reset/PE reset clears flags and accumulators before each operation. The RAM wrapper remains unchanged.

Verification. Both independent matrix cases pass all 16 C RAM byte comparisons. The clean/exception/clean sequence passes all 48 comparisons and checks reset clearing. No Part I exhaustive regression was repeated. Source, testbench, Makefile, matrices, raw reports and waveform databases are packaged separately.

Core-only synthesis metric	Integer	FP8
Cell area (library area units)	19,110.834	33,263.045
Worst full-interface setup slack (ns)	5.092169	4.936644
Estimated setup-limited period (ns)	4.907831	5.063356
Estimated full-interface frequency (MHz)	203.76	197.50
Register-to-register estimate (MHz)	268.88	254.42

FP8 cell area is 1.741x integer (+74.1%). Frequency is derived as 1000 / (10 - setup slack), with fixed constraints. These are pre-layout estimates, not measured routed operating frequencies.

Test matrices and expected results

E3M4, bias 3; round-to-nearest-even; DAZ/FTZ; separate rounding for multiply and add

Case 1: clean, flags 00000 | Entries show decimal value (hex encoding).

A row	0	1	2	3
0	1 (30)	1.5 (38)	-1.5 (B8)	-1 (B0)
1	-1.5 (B8)	-0.5 (A0)	0.5 (20)	0.5 (20)
2	1 (30)	1 (30)	1 (30)	-0.5 (A0)
3	1 (30)	-0.5 (A0)	-1.5 (B8)	1 (30)

B row	0	1	2	3
0	0.5 (20)	1.5 (38)	0.5 (20)	-0.5 (A0)
1	-1.5 (B8)	-1 (B0)	0.5 (20)	1 (30)
2	-0.5 (A0)	-1 (B0)	-1 (B0)	-0.5 (A0)
3	-0.5 (A0)	1.5 (38)	1 (30)	1.5 (38)

C row	0	1	2	3
0	-0.5 (A0)	0 (00)	1.75 (3C)	0.25 (10)
1	-0.5 (A0)	-1.5 (B8)	-1 (B0)	0.75 (28)
2	-1.25 (B4)	-1.25 (B4)	-0.5 (A0)	-0.75 (A8)
3	1.5 (38)	5 (54)	2.75 (46)	1.25 (34)

Case 2: finite overflow, flags 10001 | Entries show decimal value (hex encoding).

A row	0	1	2	3
0	1 (30)	1.5 (38)	-1.5 (B8)	-1 (B0)
1	-1.5 (B8)	-0.5 (A0)	0.5 (20)	0.5 (20)
2	1 (30)	1 (30)	1 (30)	-0.5 (A0)
3	1 (30)	-0.5 (A0)	-1.5 (B8)	8 (60)

B row	0	1	2	3
0	0.5 (20)	1.5 (38)	0.5 (20)	-0.5 (A0)
1	-1.5 (B8)	-1 (B0)	0.5 (20)	1 (30)
2	-0.5 (A0)	-1 (B0)	-1 (B0)	-0.5 (A0)
3	-0.5 (A0)	1.5 (38)	1 (30)	2 (40)

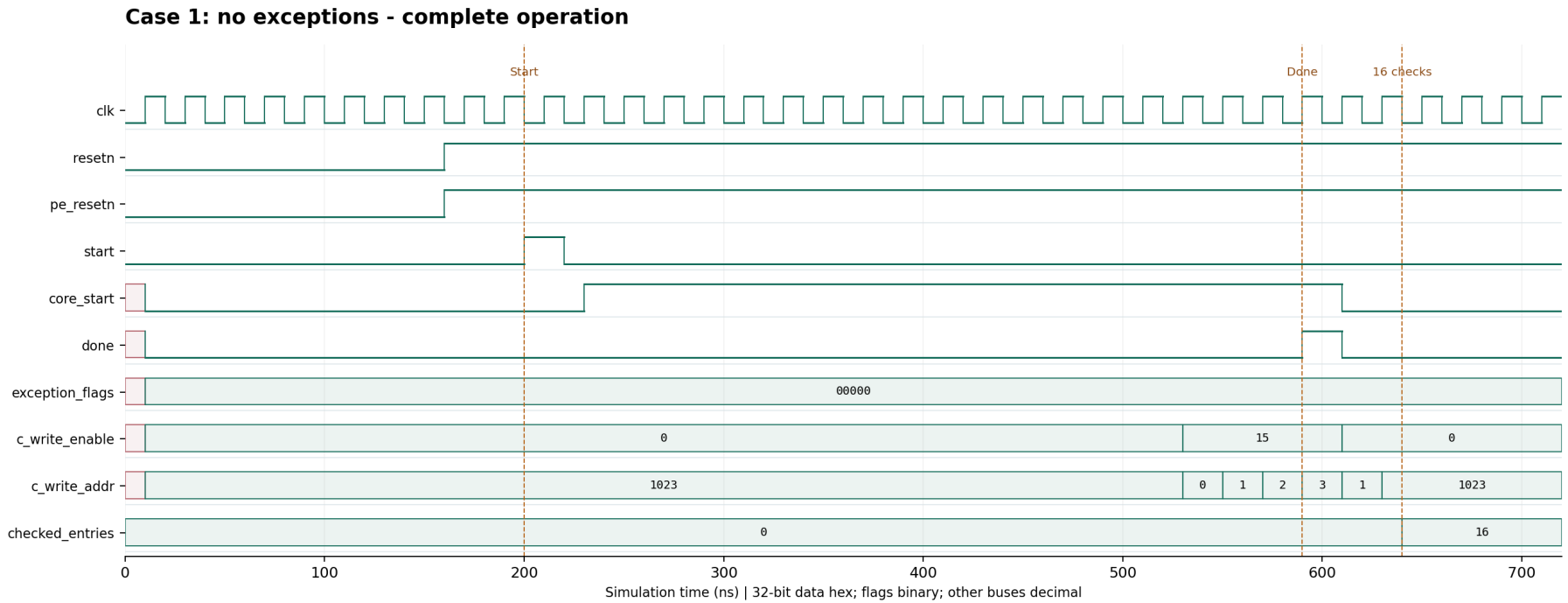
C row	0	1	2	3
0	-0.5 (A0)	0 (00)	1.75 (3C)	-0.25 (90)
1	-0.5 (A0)	-1.5 (B8)	-1 (B0)	1 (30)
2	-1.25 (B4)	-1.25 (B4)	-0.5 (A0)	-1 (B0)
3	-2 (C0)	15.5 (6F)	10 (64)	+Inf (70)

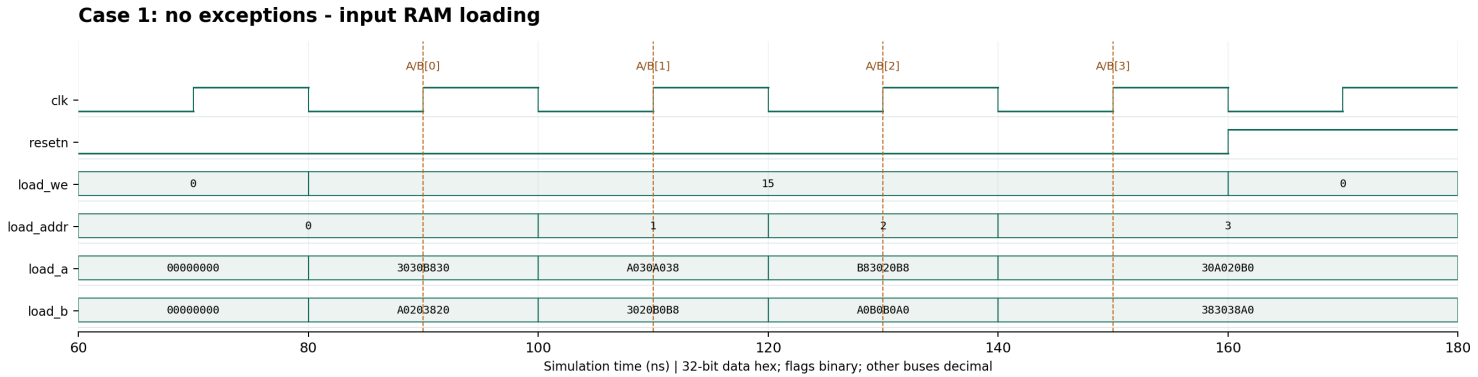
RAM packing: A word k holds column k; B word k holds row k; C word k holds column k. Element 0 occupies bits [7:0]. Clean operands are varied signed values selected with seed 32026. Case 2 changes only A[3][3] to 8 (60) and B[3][3] to 2 (40). Expected results use the supplied Part I exact-rational model.

The FP8 range ends at 15.5. PE33 multiplies 8 x 2, producing +Inf (70) with overflow and inexact. C[3][2] also rounds 9.75 to 10. The final combined exception value is 10001.

Case 1: no exceptions - load and start

Annotated plots of actual VCS VCD transitions; matching FSDB files are included in the evidence archive

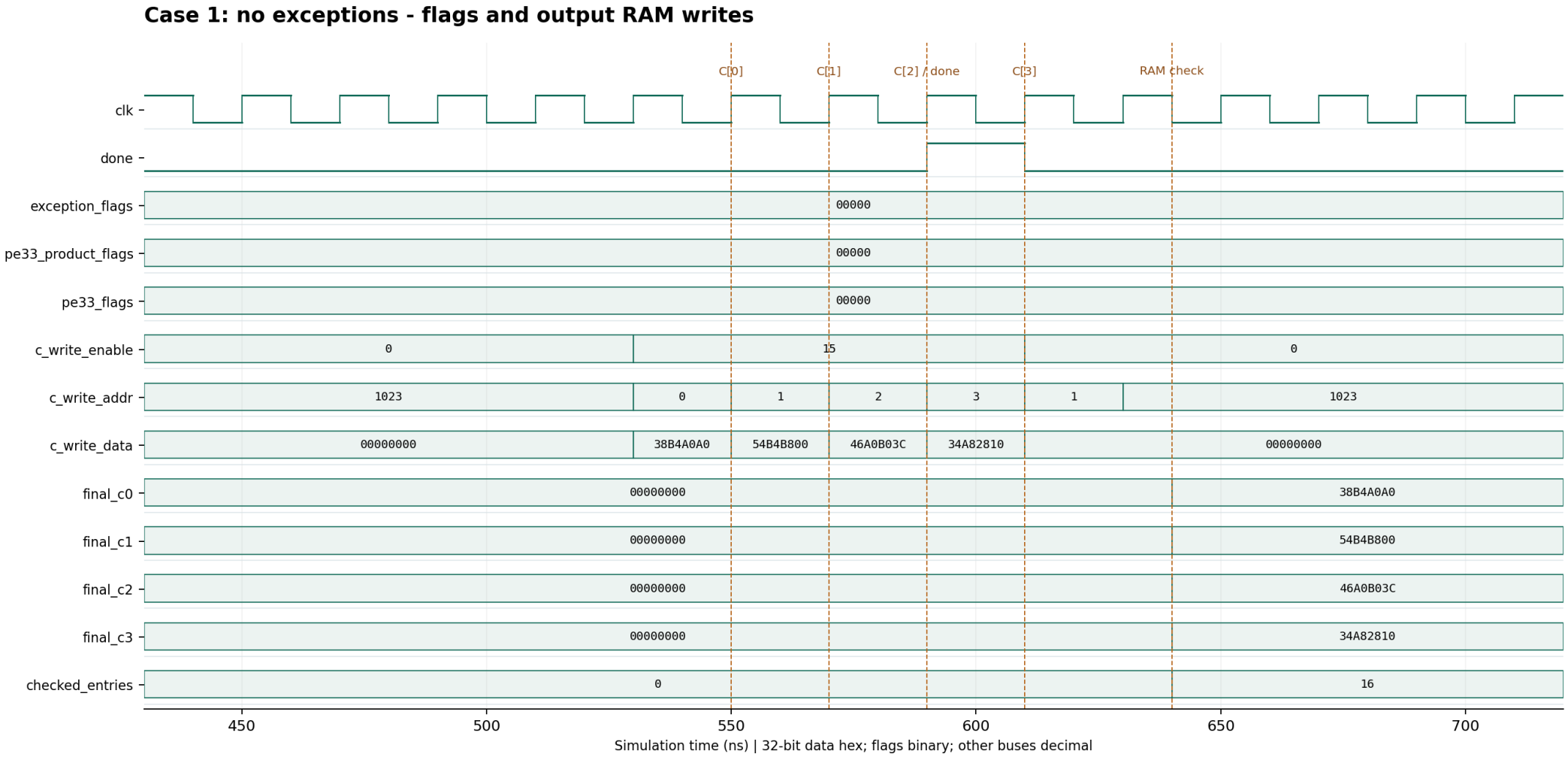




A/B data is driven at 80, 100, 120 and 140 ns, then stored on the rising clock edges at 90, 110, 130 and 150 ns. Reset stays asserted during loading. External start pulses at 200 ns. Initial X values before the first reset clock are visible in the complete trace.

Case 1: no exceptions - results

C RAM writes, sticky flags and actual RAM snapshots after all 16 output comparisons

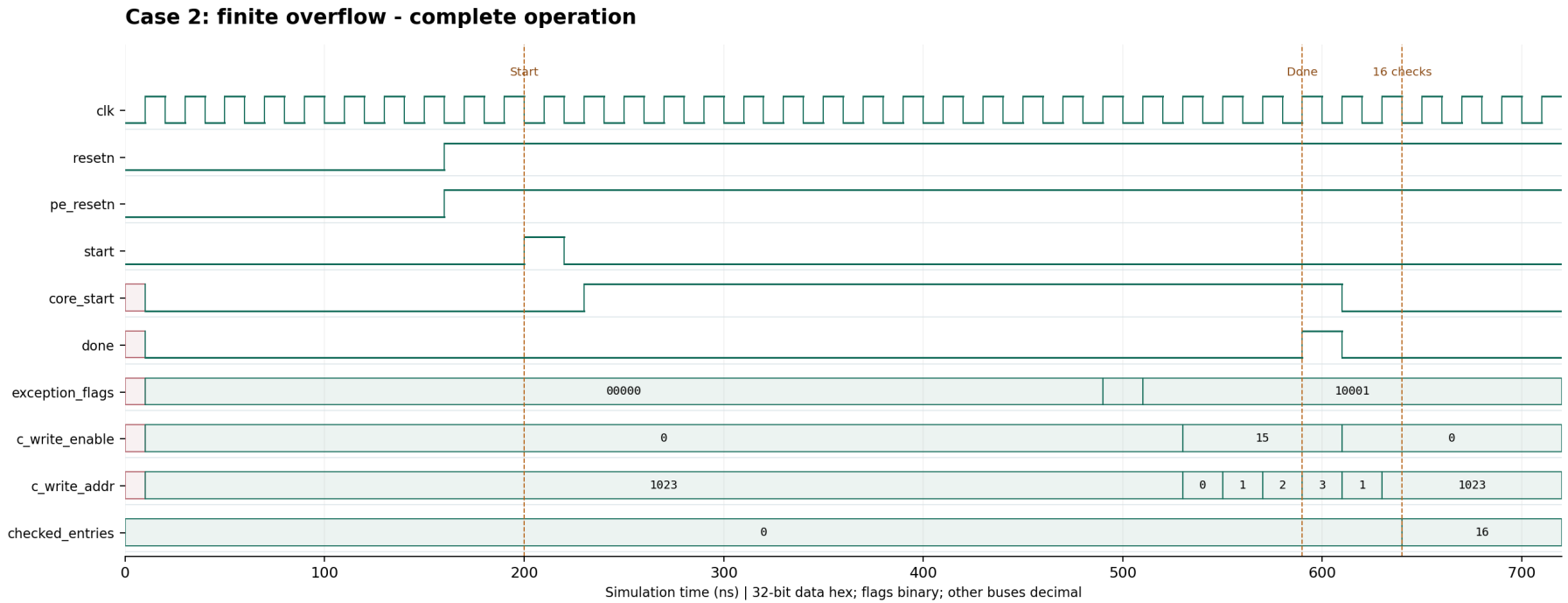


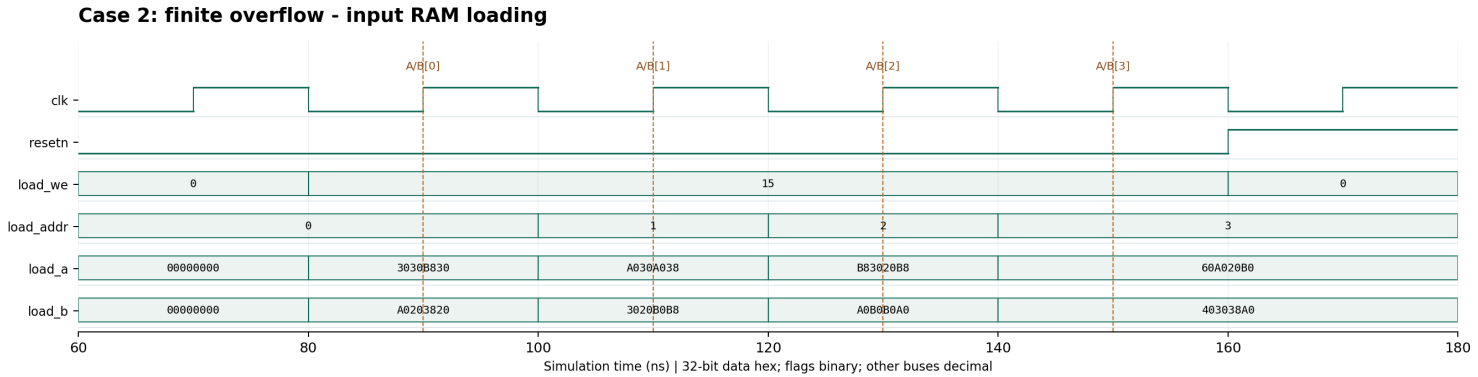
C words 0-3 are written at 550, 570, 590 and 610 ns. Done asserts at 590 ns; the final write is one clock later. Each write uses the bus value immediately before the rising edge. final_c0..3 are actual RAM snapshots captured at 640 ns after comparison, not the expected-value array.

Flags remain 00000 throughout the clean operation. The testbench reports PASS for all 16 entries at 640 ns.

Case 2: finite overflow - load and start

Annotated plots of actual VCS VCD transitions; matching FSDB files are included in the evidence archive

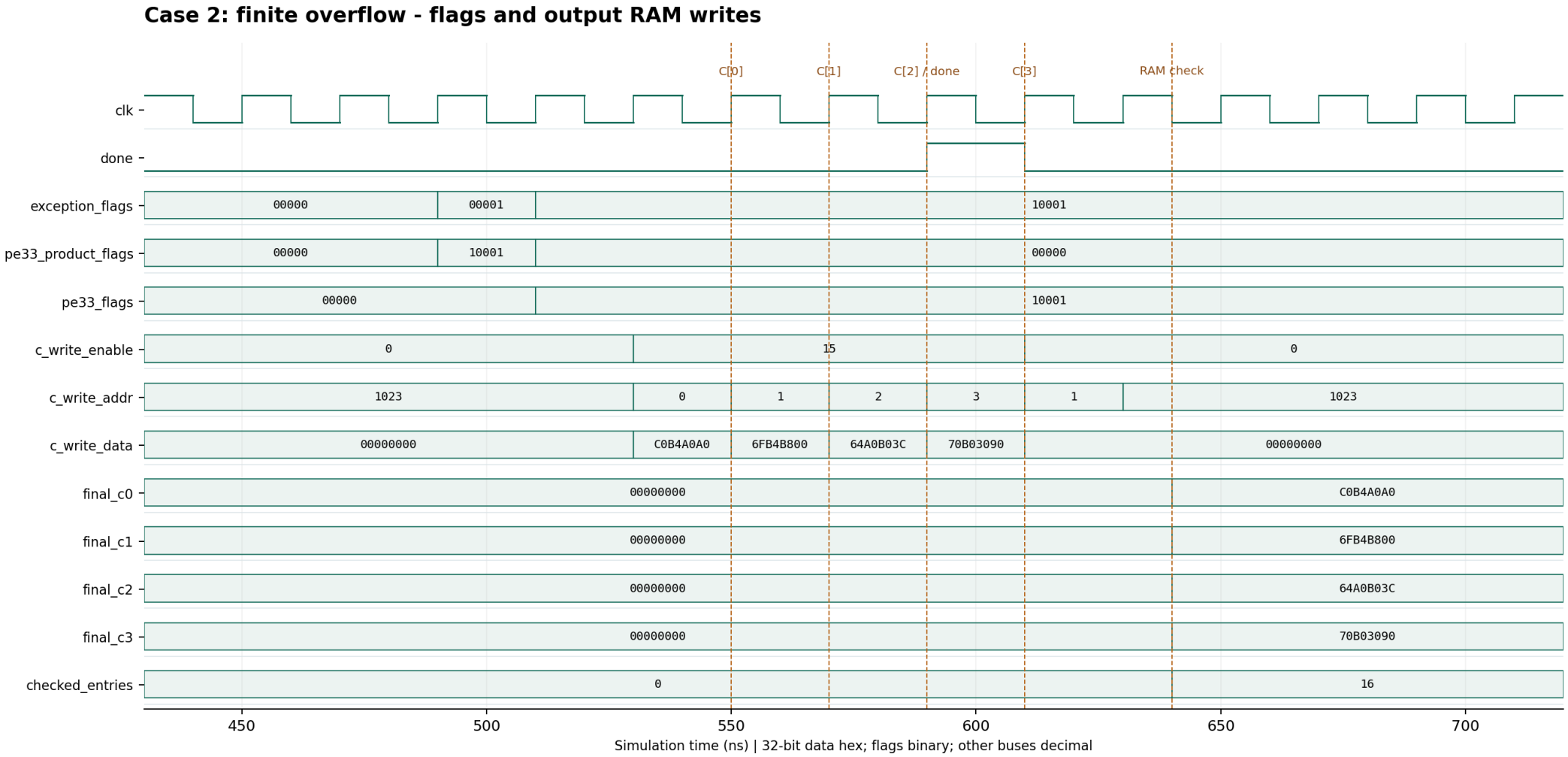




A/B data is driven at 80, 100, 120 and 140 ns, then stored on the rising clock edges at 90, 110, 130 and 150 ns. Reset stays asserted during loading. External start pulses at 200 ns. Initial X values before the first reset clock are visible in the complete trace.

Case 2: finite overflow - results

C RAM writes, sticky flags and actual RAM snapshots after all 16 output comparisons



C words 0-3 are written at 550, 570, 590 and 610 ns. Done asserts at 590 ns; the final write is one clock later. Each write uses the bus value immediately before the rising edge. final_c0..3 are actual RAM snapshots captured at 640 ns after comparison, not the expected-value array.

Combined flags become 00001 at 490 ns and 10001 at 510 ns. PE33 product flags are 10001 from 490-510 ns; the MAC sticky flag retains them afterward. The final C[3][3] byte is 70 (+Inf). All 16 comparisons pass with the expected flags at 640 ns.

Synthesis conditions and reproduction

Equal constraints on integer and FP8 cores; source and evidence manifest

Tools and scope. VCS W-2024.09-SP2-3; Design Compiler V-2023.12-SP5-1. Only matmul_4x4_systolic is synthesized. RAMs and the wrapper are excluded. Both designs retain their full public interfaces with no case-analysis specialization. Library: gscl45nm.db under the course FreePDK45 osu_soc library, typical corner. Both use compile_ultra.

Constraint	Value
Clock period / frequency target	10 ns / 100 MHz
Clock uncertainty / clock transition	0.10 ns / 0.10 ns
Input delay / output delay	1 ns / 1 ns
Input transition / output load	0.10 ns / 0.01 pF
Simulation clock period	20 ns

Timing interpretation and limitations. Worst full-interface setup paths start at final_mat_mul_size[1] and end at the output address register. The register-only estimate separately uses reg_to_reg.rpt. Reports show zero setup violations, but retain hold violations (integer 556; FP8 461, worst approximately -0.064 ns) and maximum-capacitance violations (integer 1294; FP8 3657). No wire-load model, placement, clock-tree synthesis or routing is included. These results compare mapped cores; they do not establish physical-design closure.

Reproduce. From the repository root: source env.cshrc in tcsh, cd part2_matmul4, then make fsdb and make synth-integer synth-fp8. Open build/clean.fsdb or build/exception.fsdb in Verdi. make vectors regenerates matrices; python tools/plot_waveforms.py redraws the included annotated PNGs from the VCD files (matplotlib required). See README.md for details.

Files. Source ZIP: unchanged required baseline/Part I dependencies, Part II RTL, testbench, Makefile, synthesis script, matrices, report, PNGs and raw reports. Evidence ZIP: reports plus clean/exception/sequence VCD and FSDB files. No simulator binaries or synthesis databases are needed in the source ZIP.

Source references. Rahul-Jula/Systolic-Matrix-Multiplier, base commit 808743320beff4b5da859ceec90c5731d636b920; course Lab 3 Part II handout at adventlab.notion.site; included reports/*_sim.log and reports/{integer,fp8}/*.rpt. Part III and Part IV are not implemented.